

**Title:** Git Hands-On Lab 3 - Branching and Merging**Problem Statement:****Objectives:**

- Explain branching and merging
- Explain about creating a branch request in GitLab
- Explain about creating a merge request in GitLab
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In this hands-on lab, you I learnt how to:

- Construct a branch, do some changes in the branch, and merge it with master (or trunk)

**Prerequisites:**

- Setting up Git environment with P4Merge tool for Windows

**Branching:**

1. Create a new branch “GitNewBranch”. Command: git branch GitNewBranch

```
MINGW32/c/Users/pchai/GitDemo
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git ~/GitDemo
git: 'C:/Users/pchai/GitDemo' is not a git command. See 'git --help'.

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ cd ~/GitDemo

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git branch GitNewBranch
```

2. List all the local and remote branches available in the current trunk. Observe the “\*” mark which denote the current pointing branch. Command: git branch -a

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git branch -a
  GitNewBranch
* master
  remotes/origin/main
  remotes/origin/master
```

3. Switch to the newly created branch. Add some files to it with some contents. Commands: git checkout GitNewBranch; echo "This is content in GitNewBranch written by chaitanya\_Pyla" > branchfile.txt

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ echo "This is contetent in GitNewBranch written by chaitanya_pyla" > branch
file.txt
```

4. Commit the changes to the branch. Commands: git add branchfile.txt; git commit -m "Add branchfile.txt in GitNewBranch"

```
pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ git add branchfile.txt
warning: in the working copy of 'branchfile.txt', LF will be replaced by CRLF
the next time Git touches it

pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ git commit -m "Add branchfile.txt in GitNewBranch"
[GitNewBranch c79fefc] Add branchfile.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branchfile.txt
```

5. Check the status with “git status” command. Command: git status  
I have done this step.

```
pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ git push -u origin GitNewBranch
```

**Merging:**

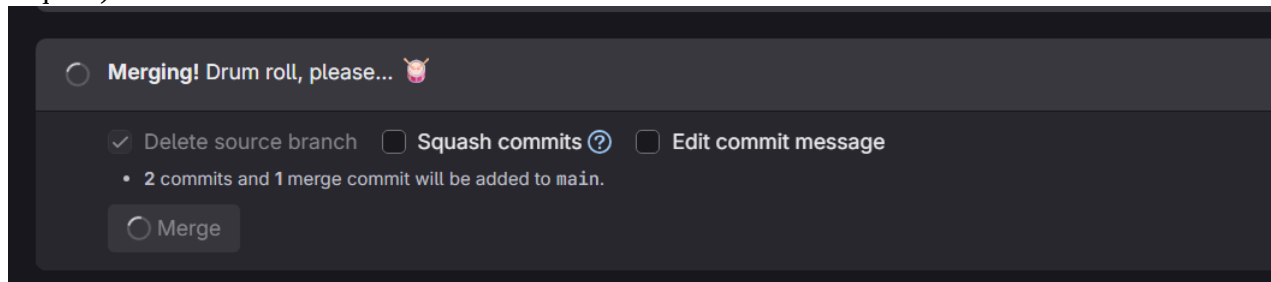
1. **Switch to the master.** Command: `git checkout master`

```
pchai@Chaitanya MINGW32 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
```

2. List out all the visual differences between master and branch using P4Merge tool. Command:

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
```

3. Merge the source branch to the trunk. Command: `git merge GitNewBranch` (or via GitLab merge request)



4. Observe the logging after merging using “`git log --oneline --graph --decorate`” Command: `git log --oneline --graph --decorate`

```
pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git pull origin main
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 275 bytes | 91.00 KiB/s, done.
From https://gitlab.com/chaitanya5850432/GitDemo
* branch      main      -> FETCH_HEAD
   452af07..797a5b0  main      -> origin/main
Updating 22be986..797a5b0
Fast-forward
 branchfile.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branchfile.txt
```

5. Delete the branch after merging with the trunk and observe the git status. Command: `git branch -d GitNewBranch`; `git status`

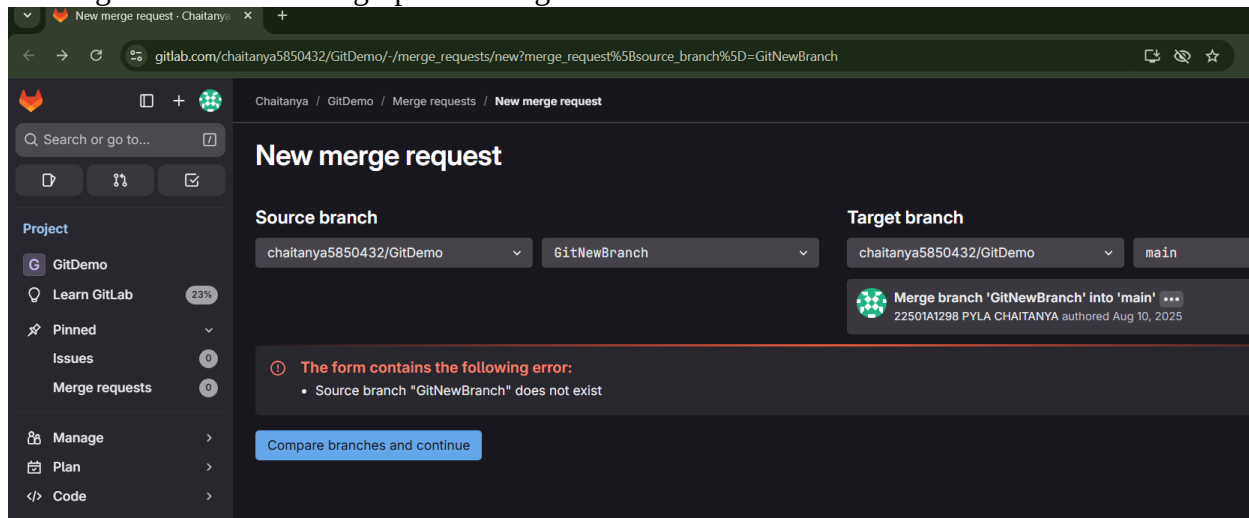
| Name           | Last commit                         | Last update    |
|----------------|-------------------------------------|----------------|
| .gitignore     | Ass .gitignore to ignore .log files | 52 minutes ago |
| README.md      | Initial commit                      | 8 hours ago    |
| branchfile.txt | Add branchfile.txt in GitNewBranch  | 16 minutes ago |
| welcome.txt    | Add welcome.txt                     | 8 hours ago    |

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### 3. Git-HOL

#### Additional Steps Performed (Based on Screenshots):

- Pushing branch to remote: `git push -u origin GitNewBranch`



- Pulling updates post-merge: `git pull origin main`

